Midterm 1 study guide

How to prepare — Lectures 01–14 and Quizzes 1–4

Last updated: October 12, 2026.

Exam: Friday, October 2 · in class (exam not posted online) · day page. Practice problems: Practice packet. Schedule: Schedule.

Recommended study plan

1. **Scan the learning outcomes below.** Mark each as: confident / shaky / blank.
2. **Go through the lecture notes (01–14) and Lab 0.** For each outcome you marked shaky or blank, find the matching fill-in, worked example, or Check yourself item. (Graded Labs 1–6 start after Midterm 1.)
3. **Build your own "important definitions and formulas" list.** As you read, write down every definition or formula you might need on the exam.
4. **Rework examples from lecture and quizzes.** Cover the solution, redo Warm-up / Worked / Check yourself / quiz-prep items by hand, then uncover and check using the filled in notes in Canvas.
5. **Start memorizing your list of definitions and formulas**.
6. **Try the practice midterm problems with no notes.
7. **Check answers** (end of the practice packet). For every miss, update your definitions/formulas list and redo a similar lecture example.
8. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

Useful Course materials

Source	Role
Lectures 01–14	Core ideas and skills
Quizzes 1–4	Fluency checkpoints (rework those styles)
Practice packet	Definitions, easy computations, harder items
Boyd & Vandenberghe, *VMLS*	Textbook (free online)

Learning outcomes (Lectures 01–14)

Lecture 01 — Vectors as representations

1. Identify objects that can be represented as vectors.
2. Interpret vector entries, units, and indexing in context.
3. Distinguish scalars from vectors.
4. Translate a small data, position, signal, or measurement problem into vector form.

Lecture 02 — Vector arithmetic and geometry

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in $\mathbb{R}^2$ and recognize corresponding operations in $\mathbb{R}^3$.
4. Check whether an answer has sensible dimensions and units.

Lecture 03 — Linear combinations (+ Lab 0)

1. Form and interpret linear combinations.
2. Explain the meaning of coefficients in context.
3. Enter, index, and modify vectors in Octave or Python.
4. Perform vector arithmetic and create a basic plot.
5. Modify supplied code and explain its output or error message.

Lecture 04 — Inner product

1. Compute a dot product.
2. Interpret it as a weighted sum.
3. Interpret its sign and magnitude as alignment.
4. Recognize orthogonal vectors from a zero dot product.

Lecture 05 — Linear and affine functions

1. Distinguish linear from affine functions in examples.
2. Test superposition numerically on simple functions.
3. Construct a function $a^\mathsf{T}x + b$ for a contextual quantity.
4. Evaluate a model by hand and computationally.

Lecture 06 — Linear functions in applications

1. Model a score, cost, prediction, or aggregate quantity.
2. Interpret model coefficients and intercepts.
3. Compare outputs from competing models.
4. State a reasonable limitation or assumption of a linear model.

Lecture 07 — Norm, distance, RMS, and variation

1. Compute vector norms and distances.
2. Interpret norm and distance geometrically and contextually.
3. Relate norm to RMS and, where appropriate, variation around an average.
4. Use software to calculate distances in dimensions too large for hand computation.

Lecture 08 — Angle and similarity

1. Compute cosine similarity or an angle between vectors.
2. Distinguish closeness in position from similarity in direction.
3. Choose an appropriate similarity measure for a stated purpose.
4. Explain why normalization may change a comparison.

Lecture 09 — Nearest neighbors

1. Formulate a nearest-neighbor question using vectors.
2. Find a nearest neighbor in a small dataset by hand.
3. Explain how feature scale can affect the answer.
4. Justify a choice of distance or similarity measure.

Lecture 10 — Clustering and k-means

1. Explain the goal of clustering.
2. Compute one assignment/update iteration of k-means.
3. Interpret a centroid as a vector.
4. Recognize that clustering output depends on representation, scaling, and initialization.

Lecture 12 — Linear dependence as redundancy

1. Recognize simple dependent and independent collections.
2. Construct a dependence relationship.
3. Interpret dependence as redundant information.
4. Explain why redundancy matters in data, measurement, and computation.

Lecture 13 — Span, independence, and subspaces

1. Define span using linear combinations.
2. Determine whether a vector lies in a simple span.
3. Interpret spans geometrically as lines, planes, or higher-dimensional collections.
4. Recognize basic subspaces and distinguish them from shifted lines or planes.

Lecture 14 — Basis, dimension, and orthonormal vectors

1. Explain what a basis accomplishes.
2. Identify a basis and dimension in simple examples.
3. Use linear combination, span, subspace, independence, basis, and dimension accurately.
4. Recognize an orthonormal collection.
5. Explain why orthonormal coordinates are convenient for measurement and reconstruction.